EXPERIMENT 1: MATLAB CODE

Title: Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

AIM: To calculate the cutoff frequency and propagation constant of TE_{10} and TM_{11} modes in a WR-90 rectangular waveguide and plot their propagation characteristics.

MATLAB program

```
clc;
clear;
close all;
% Experiment 1: TE and TM Modes in Rectangular Waveguide
c = 3e8;              % Speed of light (m/s)
a = 22.86e-3;         % Waveguide width (m)
b = 10.16e-3;         % Waveguide height (m)
L = 100e-3;           % Waveguide length (m)
f = 10e9;             % Operating frequency (Hz)
% Cutoff frequencies
fc_TE10 = c/(2*a);
m = 1;
n = 1;
fc_TM11 = (c/2)*sqrt((m/a)^2 + (n/b)^2);
% Propagation constant
beta_TE10 = (2*pi*f/c)*sqrt(1-(fc_TE10/f)^2);
if f > fc_TM11
   beta_TM11 = (2*pi*f/c)*sqrt(1-(fc_TM11/f)^2);
else
   beta_TM11 = 0;
end
% Display calculated results
fprintf('Operating Frequency = %.2f GHz\n',f/1e9);
```

```
fprintf('TE10 Cutoff Frequency = %.3f GHz\n',fc_TE10/1e9);
fprintf('TM11 Cutoff Frequency = %.3f GHz\n',fc_TM11/1e9);
fprintf('TE10 Propagation Constant = %.2f rad/m\n',beta_TE10);
fprintf('TM11 Propagation Constant = %.2f rad/m\n',beta_TM11);
% Frequency sweep
freq = linspace(5e9,20e9,500);
beta1 = zeros(size(freq));
beta2 = zeros(size(freq));
for k = 1:length(freq)
    if freq(k) > fc_TE10
        beta1(k) = (2*pi*freq(k)/c)* ...
            sqrt(1-(fc_TE10/freq(k))^2);
    end
 if freq(k) > fc_TM11
        beta2(k) = (2*pi*freq(k)/c)* ...
            sqrt(1-(fc_TM11/freq(k))^2);
    end
end

% Plot propagation constant
figure;
plot(freq/1e9,beta1,'b','LineWidth',2);
hold on;
plot(freq/1e9,beta2,'r','LineWidth',2);
xline(fc_TE10/1e9,'--b','TE10 cutoff');
xline(fc_TM11/1e9,'--r','TM11 cutoff');
xlabel('Frequency (GHz)');
ylabel('Propagation Constant \beta (rad/m)');
title('Propagation Characteristics of TE10 and TM11 Modes');
```

```
legend('TE10','TM11','Location','northwest');
grid on;
hold off;
% Illustrative TE10 electric-field distribution
x = linspace(0,a,100);
y = linspace(0,b,50);
[X,Y] = meshgrid(x,y);
Ey = sin(pi*X/a);
figure;
surf(X*1000,Y*1000,Ey);
xlabel('Width x (mm)');
ylabel('Height y (mm)');
zlabel('Normalized E-field');
title('Illustrative TE10 Electric-Field Distribution');
shading interp;
colorbar;
```

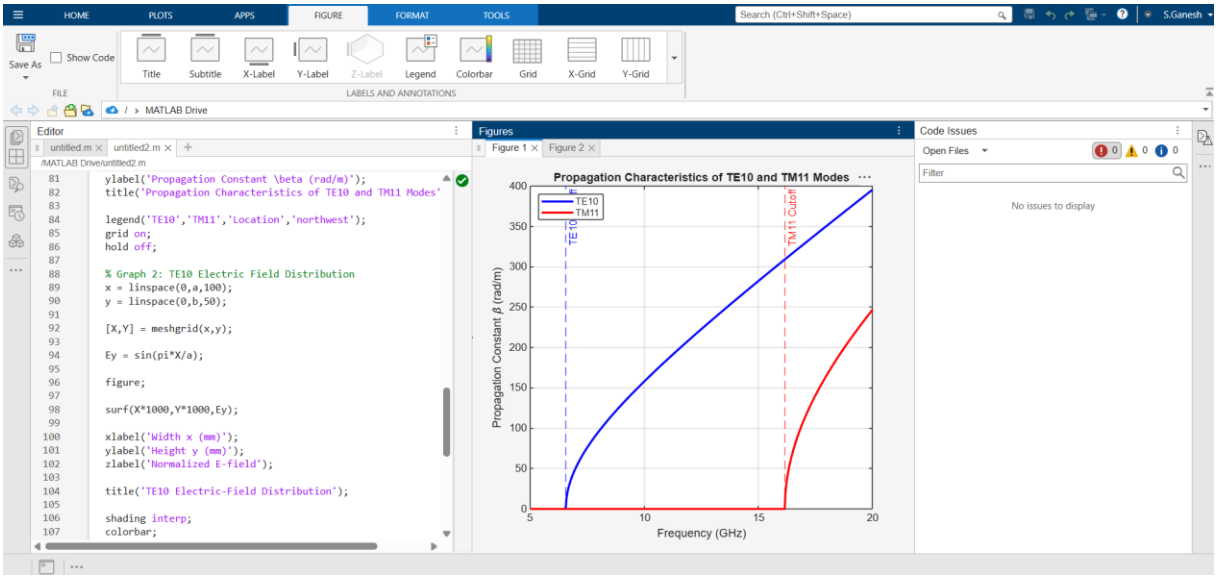
Expected output (calculated)

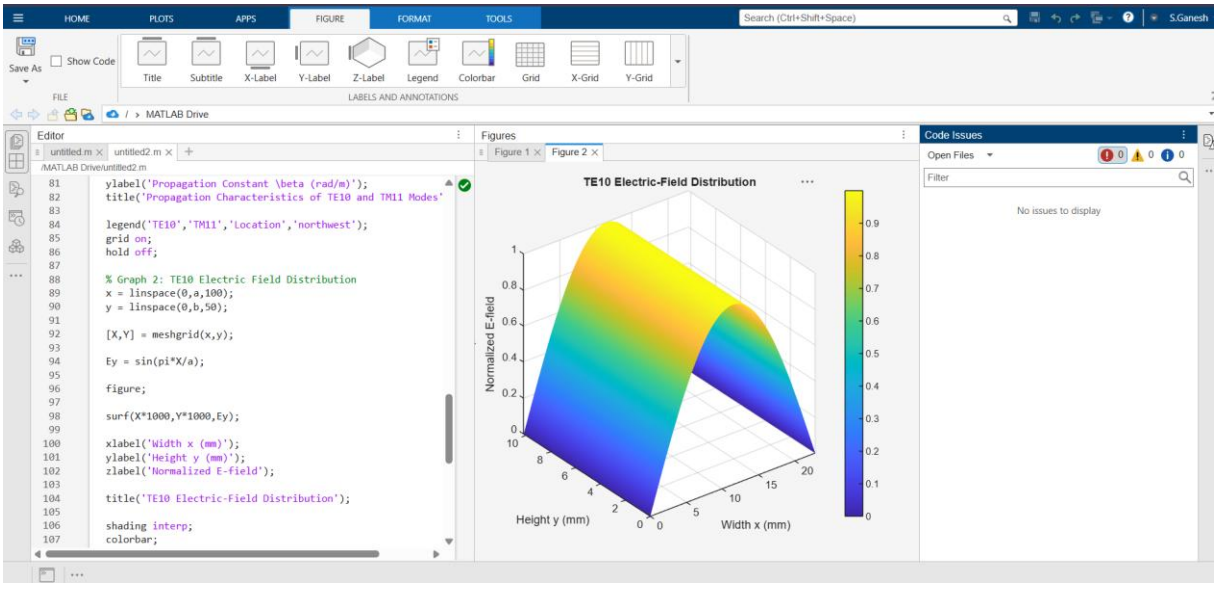
Illustrative theoretical values; not measured HFSS results.

Parameter	**Expected value**
Operating frequency	10 GHz
TE_{10} cutoff frequency	6.562 GHz
TM_{11} cutoff frequency	16.42 GHz
TE_{10} propagation constant at 10 GHz	Approximately 158.2 rad/m
TM_{11} at 10 GHz	Below cutoff; no real propagating β

RESULT:

The cutoff frequencies and propagation characteristics of the TE_{10} and TM_{11} modes were calculated using MATLAB. At 10 GHz, the dominant TE_{10} mode propagates, whereas the TM_{11} mode is below cutoff.


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81 ylabel('Propagation Constant \beta (rad/m)');
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84 legend('TE10','TM11','Location','northwest');
85 grid on;
86 hold off;
87
88 % Graph 2: TE10 Electric Field Distribution
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91
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93
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95
96 figure;
97
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99
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101 ylabel('Height y (mm)');
102 zlabel('Normalized E-field');
103
104 title('TE10 Electric-Field Distribution');
105
106 shading interp;
107 colorbar;
Figures
Figure 1
Figure 2
Propagation Characteristics of TE10 and TM11 Modes
TE10
TM11
TE10 Cutoff
TM11 Cutoff
Propagation Constant β (rad/m)
Frequency (GHz)
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Figure 2
TE10 Electric-Field Distribution
Normalized E-field
Height y (mm)
Width x (mm)
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